

A Nail-tip Device for Gesture and Force Input

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Figure 1: (a) the prototype device with three touch sensors and a strain gauge to be mounted on the fingernail tip. (b) the proposed technique in use. The thumb applies force to the tip of the finger mounting the device. Five gestures and four force inputs are supported. (c) an application example with multiple fingers. The user can scroll the map using the middle finger and zoom in/out the ring finger. (d) a confusion matrix showing the results of the gesture recognition experiment with six participants.

ABSTRACT

Micro-gestures are critical to expanding the personal computing environment. A nail-mounted device enabling micro-gesture and force inputs via the tip of a fingernail is proposed. A prototype with three touch sensors and a strain gauge to support five gestures and directional force input is fabricated. Experiments confirm successful gesture recognition. We introduce three applications to discuss the design space of the proposal.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Nail-mounted, Wearable, Gesture, Force Input

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1 INTRODUCTION

The computing environment is continuing to migrate from within the office or home to anytime and anywhere thanks to advanced portable or wearable devices, such as smartphones, smartwatches, and smart glasses. Micro-gestures have been attracting attention in order to support various environments, since they can hide the user's intention from others and decrease social disturbance [1]. As micro-gestures are typically made by bringing the pads of two fingers together, it is often necessary to attach sensors to the finger pads [4], which interfere with daily life. Fortunately, micro-gestures can be captured with only minor life-style interference by mounting devices on the fingernails.

For example, Nail+ [2] puts a device consisting of nine strain gauges on the user's fingernail to detect the force applied to the finger. It recognizes several gestures, e.g. left and force-applied left, conducted by the finger on a rigid surface. NailO [3] uses a thumb-fingernail-mounted self-contained device which includes a battery, circuit, and touch pad. The user touches the device with their index or middle finger to input. Since unintentional touch or force occurs all too easily in everyday life, challenges with these finger-nail mounted devices include how to prevent the Midas touch problem and a limited input vocabulary.

2 PROPOSED METHOD

We propose a fingernail-mounted device that combines touch and force, which indicates that it encompasses both [2] and [3]. The unique difference from conventional techniques is that the device does not cover the entire fingernail but is attached to the tip of the fingernail to detect the gestures and force applied to the nail

tip. The proposed device is assumed to be mainly mounted on the middle finger and inputs are made by the thumb. It captures the gestures made by the nail tip, such as left/right swipe, and the strength of the applied force which can be used for pointer control. Mounting multiple devices on other fingers further increases the input vocabulary. It is important that force input is available only when both the touch sensor and the strain gauge have valid outputs. This configuration is expected to decrease the Midas touch problem with force input.

Figures 1 (a) and (b) show our prototype, which is attached to the user's fingernail by double-sided tape. It has three touch sensors and a strain gauge. The outputs of these sensors are captured by a micro-controller connecting to a PC. The prototype supports five gestures: left/right swipes, up/down swipes, and tap. When the thumb touches the one of the touch sensors at the edge and then the center one, a left or right swipe is recognized. When the thumb touches and releases the fingernail so as to slightly deform the fingernail, a tap is recognized. Four directional force inputs are supported as well: left/right and up/down. When force is applied, the direction of the force is determined by the sensors. For example, the left/right direction is determined by which touch sensor at the end responds, while up/down direction is given by the value of the strain gauge, positive or negative, when the center touch sensor responds. The force's strength is determined by the strain gauge's output. Though the strain gauge has a single axis, it reacts force from any direction provided the fingernail deforms at least slightly. This directional strength-sensitive force can be used as a continuous input and can be applied, for example, to realize cursor control.

3 EXPERIMENTS

Two experiments were conducted to confirm our proposal: the recognition accuracy of gesture input and the situations in which Midas touch occurs.

The first experiment involved six participants, all males in their twenties. Each participant set the device on their middle finger and performed five supported gestures with their thumb ten times in random order. The touch sensor data recorded included touch timing and duration, and the strain gauge data during the gesture. The resulting data was classified using dynamic time warping in a ten-fold cross-validation with two conditions: using all participants' data and only individual data. The results are shown in Figure 1 (d) as a confusion matrix. The average recognition accuracy was 94.6% and 93.6% using all participants' data and only individual data, respectively. This implies that there was little difference between the participants. All gestures were accurately recognized except for the right swipe of one participant who achieved only 60% accuracy.

The second experiment determined what kinds of activities yielded the Midas touch problem in daily life. One of the authors used the device and performed daily routines. We found the following activities triggered the Midas touch, opening a door, opening the cap of a plastic bottle, putting on glasses, earphones, etc., putting hands in pockets, and typing on a keyboard. It indicates that touching another finger, palm, or conductive materials can cause Midas touch. However, no Midas touch occurred when the user forcefully touched a wall or table. This implies that gestures with force could

ease the Midas touch problem. It was noted that the device rarely interfered with daily tasks.

4 APPLICATIONS

Several applications were developed to explore the design space of the proposal. The first application was as an input for a smartphone, tablet, or smart glasses. The user scrolled among the candidate selections by swipe gestures and indicated the desired one by tap. Using the proposal, the user can control a smartphone by one hand while firmly grasping it with the other hand to prevent it from being dropped. While the application currently accepts only gesture inputs, force input can be used to move the cursor for GUI applications.

The second application supports multiple devices by using different fingers. Map control including 2D panning and zooming in/out exemplifies the benefits of this configuration, see Figure 1 (c). In this application, the device set on the middle finger is assigned to pan the map with force input, and the other one set on the ring finger is used to zoom in and out with gestures. The user can adjust the panning speed by applied force and two gestures are assigned to zooming. While this application assumes that the thumb is the only finger used to apply force and make gestures, indicating the input is sequential, other finger/hand movements can be considered for different applications. For example, the user can simultaneously press the tips of the middle and ring fingers. Note that the current device supports either force or gesture input, which is set in advance. A single device would support both simultaneously by identifying them from sensor data.

The proposal assumes the coordinated action of two fingers, typically the middle finger with the device and the thumb to apply force or make gestures. It is also possible for just a single finger and any solid surface to realize input by introducing "ambient mode." The third application, a music player, was designed to explore this. The user applies force to a solid surface by the finger mounting the device for more than three seconds to activate the mode. In this mode, the user can control the music player by the single finger, for example, the volume increases when the user applies upward force. If no force is applied to the device for a certain period of time, the ambient mode is deactivated. The mode works effectively when the user cannot use their hands freely, e.g. when the user is holding onto a handrail.

5 CONCLUSION

A finger-nail mounted device that supports micro-gestures and force input was proposed. Experiments showed that gestures could be accurately recognized but further improvements are necessary because Midas touch sometimes occurs. Three applications were introduced to explore the design space.

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